Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

New grad SWE (M.S. May 2026, GPA 3.87) with **8+ years of production Python and full-stack engineering** building GenAI products and ML quality systems. Built **AgentProbe** end-to-end - a 5-metric ML quality estimation engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates, multi-provider LLM routing, and a React dashboard - the core GenAI product scale demands. Full-stack delivery across **120+ Agile sprints** at Entrata (100K+ MAU, 1.6M records); Python + TypeScript + React in production at scale. *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Python (8+ years production) | ML Quality Estimation (LLM-as-Judge) | GenAI Product Engineering | TypeScript / JavaScript / React | LLM APIs (Anthropic, OpenAI, Groq, Ollama) | RAG Pipelines + ChromaDB | FastAPI / REST APIs | Multi-Agent Systems (LangGraph) | AWS (S3, EC2, RDS, Glue) | CI/CD (GitHub Actions) | Full-Stack Delivery at Scale | SQL / PostgreSQL

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built and shipped **20+ full-stack product modules** (Python, TypeScript/JavaScript, React, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - owned features end-to-end from scoping through production.
- Applied **Python and ML pipelines** to automate data quality checks and transformation across **1.6M financial records** - production GenAI-adjacent tooling on regulated data with output validation.
- Cut microservices execution time by **50%** via Docker containerization; improved production query performance by **30%** across 200+ SQL queries; maintained **99.5% uptime** for 6+ months.
- Migrated all client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents**; built 5+ internal platform tools cutting resolution time by 40%.
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation; enforced TDD/BDD and secure coding standards across all major releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and database systems; designed supplementary exercises driving iterative quality improvement across sprint-style delivery cycles.

PROJECTS

AgentProbe: GenAI Quality Platform -- ML Evaluation Engine + Full-Stack Product OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built a GenAI quality product end-to-end: a **5-metric ML quality estimation engine** (factual accuracy, hallucination, retrieval completeness, response completeness, compliance) scoring LLM and RAG outputs with CI/CD gates that block failing runs automatically. **Multi-provider LLM routing** across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type and cost. **React dashboard** with real-time evaluation results and per-metric trend views. **15-endpoint FastAPI backend** supporting concurrent test runs; **3-agent LangGraph system** (Orchestrator, RAG Tracer, Auto Fix) with tool calls and inter-agent state handoffs. Runs 100% locally at zero cost; open-source and production-grade. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

Full-stack Flask + React product translating natural language queries to structured FHIR API calls using LLMs, BERT, and embeddings. HIPAA-compliant RBAC; **~60% reduction in manual retrieval time.** *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline -- 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML pipeline: feature engineering → model training → evaluation → deployment. **85%+ accuracy** on 100K+ records; interactive results dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · ML systems, statistical modeling, production data pipelines, experimental design · Graduating May 2026

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

GenAI & ML: LLM APIs (Anthropic, OpenAI, Groq, Ollama), LangChain, LangGraph, RAG Pipelines, ChromaDB, Vector Search, ML Quality Estimation (LLM-as-Judge), Prompt Engineering, BERT, spaCy, Hugging Face, PyTorch, scikit-learn

Full-Stack & Infra: React, FastAPI, Flask, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

Process: TDD/BDD, Integration Testing, Open-Source Development, Agile/Scrum, Git, JIRA